

Unit 13, Lesson 2: Justifying Measures of Center and Spread

Benchmark

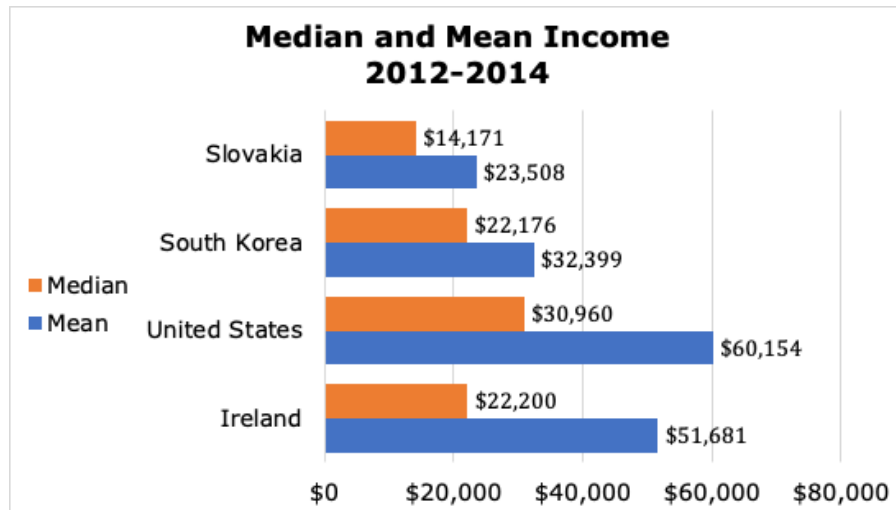
MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.

Learning Target

- ☐ I can justify whether or not a measure of center or variation appropriately summarizes data.

13.2.1 Warm-Up: Income

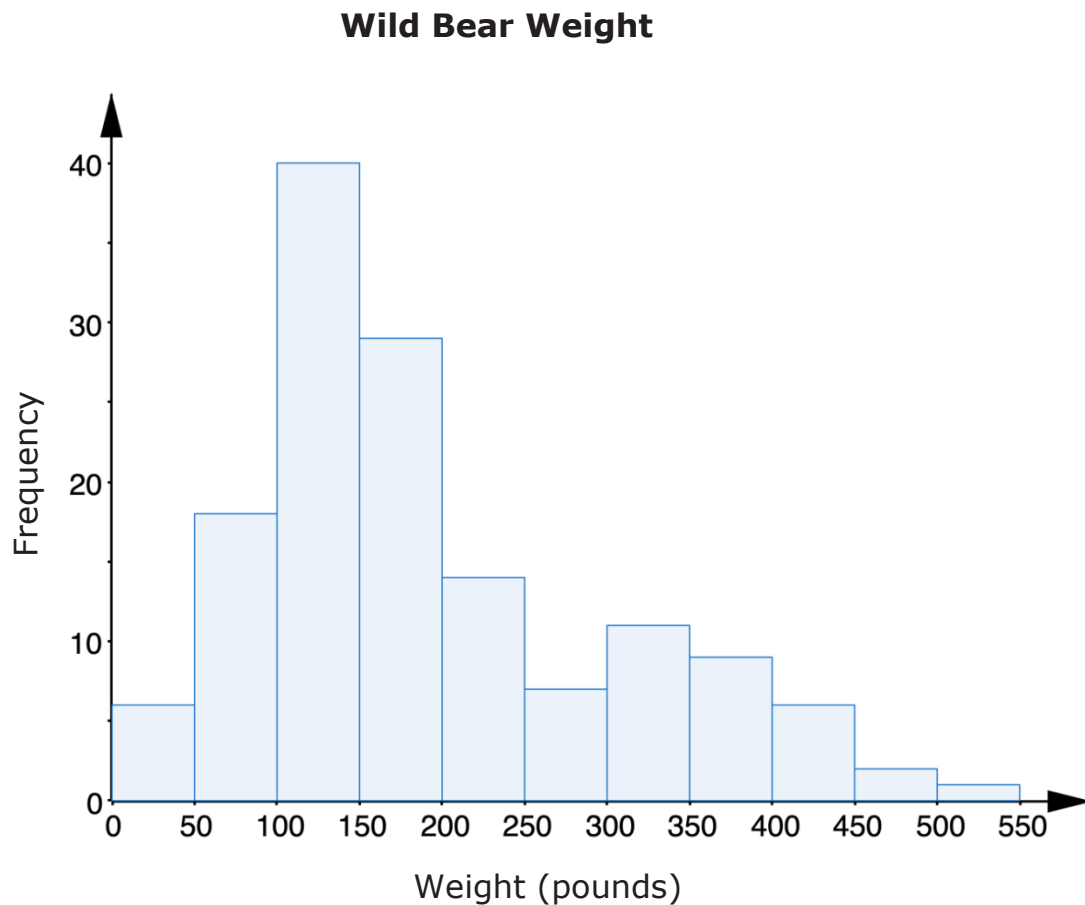
The diagram below shows bar charts for the median and mean income between 2012 and 2014 in selected countries. The income is given in U.S dollars (\$).



1. What do you notice about the values for median and mean income?
2. Why do you think the mean income for the U.S. was so much higher than the median income?

13.2.2 Collaboration: Center and Variation in Context

1. In a study of wild bears, researchers measured the weights, in pounds (lb), of 143 wild bears that ranged in age from newborn to 15 years old. The data was used to make the histogram shown.



- a. Complete the statements.

Based on the data, it appears wild bears typically weigh

between

- ☐ 50 and 150
- ☐ 100 and 200
- ☐ 150 and 250

pounds. A good estimate for the typical

weight for wild bears is _____ pounds.

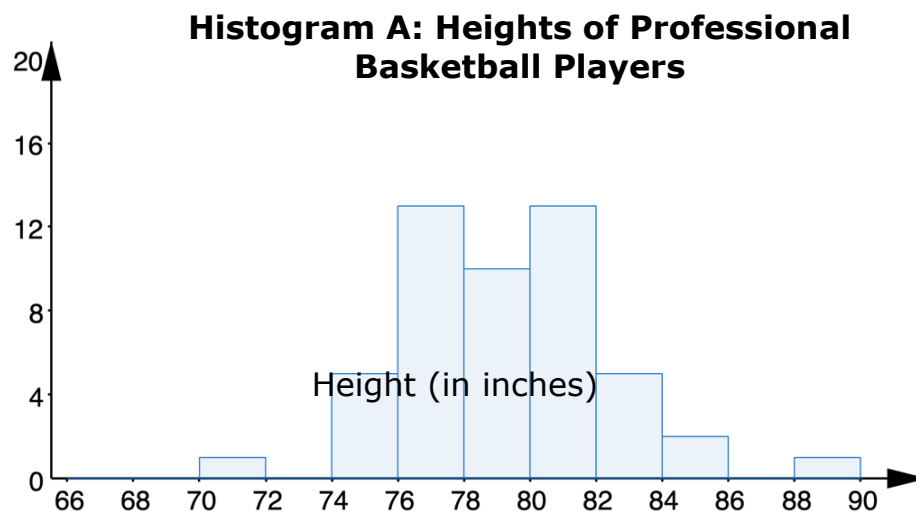
- b. Compare your estimate with your partner. Discuss whether it is possible for both of your estimates to be reasonable, even if they are not the same.
- c. Discuss with your partner whether your estimate from Part A is closer to the median or the mean. Summarize your discussion.
- d. Angelica and Raymond were asked to describe the variation in weight of the wild bears.

Angelica said, "The bear weights range from less than 50 to over 500 lbs, so the range best describes the spread of the weights of the wild bears."

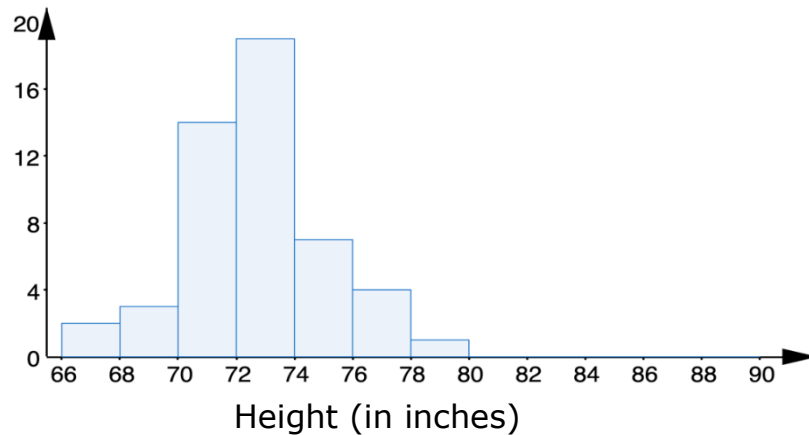
Raymond said, "It looks like most bears in the sample weigh between 100 and 300 lbs, with few weighing less and few weighing more. The small weights could be because there were babies in the sample. So, the interquartile range is the best measure of spread to describe this data."

Explain whether you agree with Angelica or Raymond. Justify your reasoning.

2. The two histograms, labeled A and B, show height distributions of 50 male professional basketball players and 50 male professional baseball players.



Histogram B: Heights of Professional Baseball Players



Work with your partner to complete each statement.

- a. In ☐ Histogram A,
☐ Histogram B, the average height is around 72 inches. The values of the mean and median are similar.
- b. To explain typical height, the variation in ☐ Histogram A
☐ Histogram B is best described by the interquartile range because the range may be misleading due to outliers. The values of the mean and median are similar.
3. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer displayed the earthworms’ lengths, in millimeters (mm), in a stem-and-leaf plot.

Stem	Leaf
0	6
1	1, 8, 9
2	0, 3, 3, 5, 5, 6, 7, 7, 8, 9
3	2, 3
4	1, 2, 8
5	2, 4, 9
6	0
7	7
8	
9	3

Key: 2|8 = 28

Use the display to answer the following questions.

- a. Half of the earthworms sampled were less than _____ mm long.
- b. The farmer reported that the average length is 35.92 mm. Why might this measure of center be misleading about the typical length of the earthworms in this sample?
- c. Compare your answers to Parts A and B with your partner and, if necessary, resolve any differences.

- d. The farmer reported that the lengths varied between 6 mm and 93 mm. The farmer's assistant said that most earthworms had lengths between 23 and 48 mm.

Work with your partner to complete the statements.

The farmer is using the

- ☐ IQR
- ☐ range

to summarize the variation in

worm length, while the farmer's assistant is using the

- ☐ IQR.
- ☐ range.

If a researcher is interested in how long

earthworms are, the

- ☐ IQR
- ☐ range

is an

appropriate measure of variation for the farmer

to report because _____

_____.

13.2.3 Bring It Together: Outlier Effects

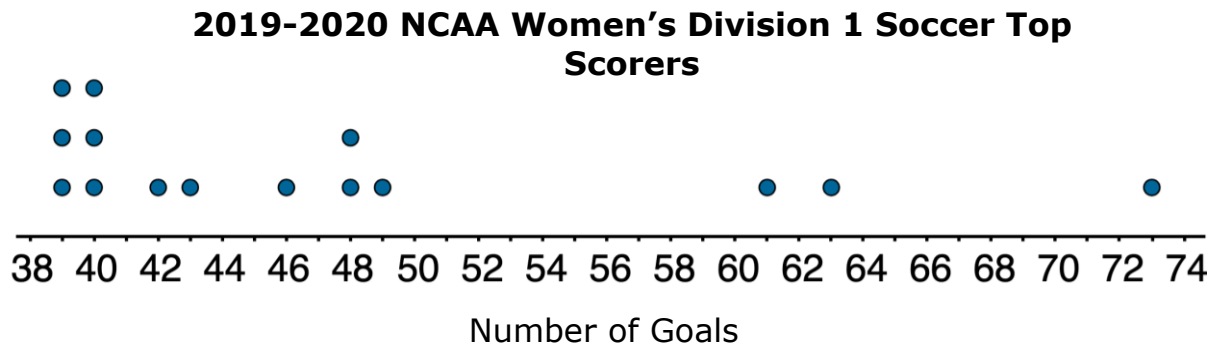
- Complete each statement to describe the effect outliers have on each measure of center or variation.
 - The presence of an outlier does not have a significant effect on the ☐ mean.
☐ median.
 - The presence of an outlier will change, often significantly, the value of the ☐ mean.
☐ median.
 - The presence of an outlier does not have a significant effect on the ☐ IQR.
☐ range.
 - The presence of an outlier will change, often significantly, the value of the ☐ IQR.
☐ range.



When outliers are present or a distribution is skewed, the _____ is the preferred measure of center and the _____ is the preferred measure of variation to describe the distribution. However, the context must be considered because outliers may be important in understanding the data.

13.2.4 Guided Practice: Comparing Distributions

1. Below is a line plot of the number of goals scored by the top 15 women goal scorers during the 2019-2020 NCAA Women's Division 1 soccer season.



Rank	Player, School	Number of Goals
1	Evelyne Viens, South Florida	73
2	Catarina Macario, Stanford	63
3	Raimee Sherle, Boise St.	61
4	Ally Watt, Texas A&M	49
5	Tziarra King, NC State	48
5	Deyna Castellanos, Florida St.	48
7	Alexa Genas, Campbell	46
8	Morgan Weaver, Washington St.	43
9	Peyton DePriest, Middle Tenn.	42
10	Elise Flake, BYU	40
10	Samantha Dewey, Xavier	40
10	Hayley Younginer, Wake Forest	40
13	Dakota Mills, Saint Joseph's	39
13	Katie McClure, Kansas	39
13	Taylor Kornieck, Colorado	39

- a. Without doing any calculations, explain which measure of central tendency you expect to be larger, the mean or the median.
 - b. Find the mean number of goals.
 - c. Find the median number of goals.
 - d. Check your work with your partner, and if necessary, resolve any differences.
 - e. Demetrica stated that the typical number of goals in the season for a scorer in the top 15 was about 48.
Explain whether you agree with Demetrica using your work from Parts A-C.
2. Demetrica described the variation by saying that most of the players in the top 15 had around the same number of goals scored, with only a few scoring much higher.
- a. Given her description, discuss with your partner whether Demetrica should use the range or the interquartile range to numerically summarize the spread of the number of goals scored.

- b. Explain which measure of variation is most appropriate to summarize the data.

3. At times, a statistical purpose may justify using a particular measure to describe the data.

Complete the statement.

If the statistical purpose of collecting the data is to determine how high the very

top scorers score, the

- ☐ IQR
- ☐ range

- ☐ IQR.
- ☐ range.

would be the most appropriate measure to use because this

information is not captured in the

13.2.5 Wrap-Up: Reflection

Draw an **X** on the line to indicate where your current learning is in relation to the following learning targets.

1. I can justify whether or not a measure of center appropriately summarizes data.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.



2. I can justify whether or not a measure of variation appropriately summarizes data.

I need help.
I can't get started.

I'm getting there, but
I need more practice.

I understand this
and I feel confident.

